

JURAJ TOMORI

Software Engineer - Volumetric Video

jtomori.github.io, learning notes

EXPERIENCE

Dimension Studio

Software Engineer, Applied Technology

May 2024 - Present

Remote

- Image, geometry processing (including Gaussian Splats), computer vision, graphics
- Pipeline development, colour management, software distribution & package management

Dimension Studio

Software Developer, RnD

September 2020 - April 2024

Remote

- Research and development for volumetric video capture and processing
- Applied Differential Rendering for material extraction from volumetric datasets (SIGGRAPH 2022)
- Implemented distributed post processing pipelines for the largest volumetric video productions: *Whitney Houston: I Wanna Dance with Somebody* (FMX 2023), *Wicked*, *Those About to Die*
- Optimized captured volumetric data storage by 50%
- Implemented an OpenColorIO-based colour management pipeline, improving visual fidelity, accuracy

Dimension Studio

Pipeline & Tools Developer

July 2019 - August 2020

London, UK

- Initiated RnD projects focused on improving assets quality for VFX, games, VR, AR and web
- Built Houdini-centric PDG-based volumetric video postprocessing pipeline
- Visualized datasets, capture stages, created synthetic data and simulations for rapid iteration

Platige Image

FX Artist

July, August 2016, August 2017

Warsaw, Poland

- FX development, shot work: Twardowsky 2.0, Frostpunk, Ghost Recon Wildlands, Skull and Bones, Crossfire HD

EDUCATION

Filmakademie Baden-Württemberg

Diploma in Technical Directing

2016 - 2019

Graduation project: VFX Fractal Toolkit (SIGGRAPH 2019)

Academy of Performing Arts in Bratislava

B.A. in Visual Effects

2013 - 2016

Thesis: Crowd simulation in SideFX Houdini

PUBLICATIONS

Reflectance Estimation for Free-viewpoint Video, Volumetric video workshop, VFX Fractal Toolkit

COURSES

HarvardX: CS50's Introduction into AI with Python, Computer Science, Web development with Python & JS

MITx: 6.00.2x, Computational Thinking with Julia, The Missing Semester of Your CS Education

UW: Programming Languages, Udacity: Interactive 3D Graphics

Dabeaz: Practical Python Programming, Advanced Python Mastery

TECHNICAL STRENGTHS

Python, Houdini, Rez, Ansible, Git, Perforce, Julia, Linux